Single-Ended Joint Estimation of HIF Location and Arc Resistance via Power-Frequency Admittance Identification with Dual-Channel Noise Modelling

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***Abstract*—High-impedance faults (HIFs) account for an estimated 5–10 % of distribution-feeder faults and roughly a quarter of downed-conductor events go undetected by conventional overcurrent protection (IEEE PSRC Working Group D15, 1996), and grid-caused ignitions represent 19 % of all U.S. wildfires over 2016–2020 (NREL TP-5R00-80746, 2023), making accurate HIF location and arc resistance estimation an urgent safety problem. This paper proposes a single-ended joint estimator of fault location α and arc resistance R_x that fits a state-space transfer function $H_{\mathrm{model}}(j\omega_0;\alpha,R_x)$ to the power-frequency single-bin DFT admittance $H_{\mathrm{meas}}(j\omega_0)$ via a two-stage optimiser (grid + multi-start, then gradient with Armijo line-search) over a 720-case sweep ($10\alpha \times 5R_x \times 4\,\mathrm{SNR}_V \times 4\,\mathrm{SNR}_I$) under independent dual-channel additive white Gaussian noise. The estimator achieves a mean fault-location error of 0.009 % in the noiseless baseline and 1.18 % at $\mathrm{SNR}_I = 20\,\mathrm{dB}^{(\text{see Glossary, LOC-ERR})}$; mean arc-resistance error is 2.42 % noiseless and ≤ 8 % across $100 – 2000\,\Omega$ at SNR $\geq 30\,\mathrm{dB}^{(\text{see Glossary, RX-ERR})}$; the location-error metric is distinct from the TF-magnitude modelling error of the two-section π-model versus a 50-section reference, which is reported separately in § II.D. The empirical RMS estimator stays within 15 % of the Cramér–Rao Lower Bound at SNR $\geq 40\,\mathrm{dB}^{(\text{see Glossary, CRLB-GAP})}$. All results use $F_s = 10$ kHz, $N_s = 200$, one cycle window$^{(\text{see Glossary, SAMP-CFG})}$. Single-frequency operation, training-free deployment, and the absence of remote-end communication make the estimator a practical single-ended primary HIF locator suitable for integration into the SAMBPS DTaaS Protection-Validation digital-twin module.**

***Index Terms*—High-impedance fault, fault location, arc resistance, transfer-function identification, single-ended, joint estimation, training-free, single-frequency, Cramér–Rao Lower Bound, dual-channel noise model, distribution protection, SAMBPS DTaaS.**

I. Introduction

High-impedance faults (HIFs) account for an estimated 5–10 % of all distribution-feeder faults, and approximately one quarter of downed-conductor incidents go undetected by conventional overcurrent relaying [1]. In the United States, grid-related ignitions caused 19 % of wildfires recorded between 2016 and 2020 [2]. Two catastrophic events frame the operational stakes: the 2009 Australian Black Saturday bushfires killed 173 people, with electrical assets implicated as ignition sources on multiple fire fronts [3]; the 2018 PG&E Camp Fire killed 85 and led to a USD 13.5 billion settlement after a worn transmission hook ignited dry grassland near Pulga, California [4].

In response, the California Public Utilities Commission (CPUC) implemented Senate Bill 901 (SB 901), placing utilities under a formal Public Safety Power Shutoff (PSPS) framework that mandates de-energisation when fire-weather risk exceeds defined thresholds [5]. PSPS shutoffs avoid catastrophic ignition but at significant social and economic cost; selective HIF location and arc-resistance estimation, by contrast, would enable targeted de-energisation of only the faulted segment, materially shrinking the PSPS footprint. The estimator developed in this paper is designed to fill that gap.

The prior-art literature on HIF location and detection clusters into seven distinct families, summarised in Table I. Two columns are added for the comparison this paper draws: "Single-ended," which marks methods that operate from a single line terminal, and "Joint α+R_x," which marks methods that estimate fault location α and arc resistance R_x *simultaneously* from the same observation rather than estimating one and treating the other as a fixed parameter or post-processing step.

The proposed estimator described in this paper would occupy an eighth row of Table I and is the only entry that is "Yes" in both the Single-ended and the Joint α+R_x columns: it operates from a single line terminal and produces a simultaneous estimate $(\hat{\alpha},\hat{R}_x)$ from a single complex admittance observation $H_{\mathrm{meas}}(j\omega_0)$.

Of the seven prior-art families, only (i), (ii), (iv), (vi) and (vii) operate from a single terminal, and none produces a simultaneous estimate of fault location and arc resistance. Families that do report R_x either treat it as a nuisance parameter solved for after location is fixed (family (ii)) or infer it implicitly through a classifier boundary (family (vi)).

Paper contributions

1) A single-ended power-frequency admittance identification formulation that derives $H_{\mathrm{meas}}(j\omega_0)$ from the dual-channel single-bin DFT and matches it against a state-space transfer function $H_{\mathrm{model}}(j\omega_0;\alpha,R_x)$ derived in closed form from the two-section π-line.
2) A two-stage optimiser (grid + multi-start, then gradient descent with Armijo line-search) that jointly estimates $(\hat{\alpha},\hat{R}_x)$ from a single complex observation, without remote-end communication and without offline training.

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TABLE I
SEVEN-FAMILY TAXONOMY OF PRIOR-ART HIF LOCATION / DETECTION METHODS. "SINGLE-ENDED" MARKS METHODS THAT USE ONLY ONE LINE TERMINAL; "JOINT $\alpha+R_x$" MARKS METHODS THAT ESTIMATE BOTH QUANTITIES SIMULTANEOUSLY FROM A SINGLE OBSERVATION. NONE OF THE SEVEN FAMILIES IS YES IN BOTH COLUMNS.

Family	**Representative anchors**	**Single-ended**	**Joint $\alpha+R_x$**	**Distinguishing trait**
(i) Signal processing / morphology / wavelet	Aucoin–Russell 1987 IEEE TPWRD [6]; Brahma 2013 IEEE TPS [7]; Mishra 2019 IEEE TII (VMD) [8]	Yes	No	Detection-only; no R_x output.
(ii) Impedance / admittance / spectral	Iurinic–Bretas 2018 IJEPES [9]; Orozco-Henao–Bretas 2020 EPSR [10]; Penaloza 2023 EPSR [11]	Yes	No	R_x enters as a parameter, not jointly estimated.
(iii) Two-ended impedance / synchrophasor	Eriksson–Saha 1985 IEEE TPAS [12]; Zeng 2021 EPSR [13]; Chandran 2024 IET GTD [14]	No	No	Requires comm channel + remote-end measurement.
(iv) Traveling-wave	SEL T401L [15]; Costa–Lopes 2015–2022 [16]; IEEE TPWRD 2024 [17]	Yes	No	MHz-bandwidth front; HIF wavefront weak / smeared.
(v) μ-PMU / data-driven	Cui–Weng 2020 IEEE TSG [18]; Shahsavari 2019 IEEE TSG [19]; Farajollahi 2018 [20]	No	No	Requires distributed μ-PMU deployment.
(vi) ML / DL / GAN / GNN	Sirojan 2022 NCAA (transformer-CNN) [21]; Veerasamy 2021 IEEE Access (LSTM) [22]; RGATv2 arXiv 2510.03571 (2025) [23]	Yes	No	Training-data dependent; opaque to operator.
(vii) Eigenvalue / state-matrix	Paramo–Bretas 2023 ISGT [24]; Bretas 2021 MDPI Appl. Sci. [25]	Yes	No	Recovers eigenstructure; R_x implicit only.

3) A 720-case study under independent dual-channel additive white Gaussian noise that reports per-cell mean and 95$^{\text{th}}$-percentile location and R_x error, including a noiseless baseline that isolates the structural floor.
4) An explicit disambiguation of two error metrics that the prior literature routinely conflates: (a) the TF-magnitude modelling error of the two-section π-model versus a 50-section reference, and (b) the location error of the estimator (Glossary entries `MODEL-ERR` and `LOC-ERR`, §VI).
5) An *identifiability bound*, derived from the Cramér–Rao inequality, that quantifies the lowest variance any unbiased single-bin admittance estimator can achieve at a given $(\alpha, R_x, \mathrm{SNR}_V, \mathrm{SNR}_I)$. We position this not as a methodological novelty but as the objective reference against which the proposed estimator's residual gap is measured (within 15 % of the Cramér–Rao Lower Bound at SNR ≥ 40 dB).

Roadmap. Section II sets up the distribution-line and arc *modelling* (two-section π-line, anti-parallel diode arc, and the section-model accuracy study); Section III presents the transfer-function *identification* (single-bin DFT cost and analytical model gradient); Section IV describes the two-stage *optimiser*; Sections V–VI carry the *validation* programme (simulation matrix, dual-channel noise model, results); Section VII compares against prior-art numerically; and Section VIII derives the Cramér–Rao identifiability *bound* that the estimator is benchmarked against. Section IX concludes with a 12 / 24 / 36-month roadmap into the SAMBPS DTaaS Protection-Validation digital-twin module.

II. SYSTEM AND MODELS

[§ II.A distribution-line parameterisation, § II.B two-section π-model state-space, § II.C anti-parallel diode arc model. WP0.5 lands the π-derivation appendix (`docs/AppendixA_derivation.pdf`); WP2.1 lands the closed-form distributed-parameter forward model

TABLE II
FORWARD-MODEL ACCURACY VS THE WP1.3 50-SECTION π REFERENCE ON THE OPERATING GRID. "MEAN" IS ACROSS ALL 95 CELLS; "MAX" IS THE WORST CELL. UPDATED UNDER WP2.3 (P2.3).

Forward model	**Mean $\lvert\Delta H\rvert/\lvert H_{\text{ref}}\rvert$**	**Max**	**So**
v1 R-L-only 2-section (no shunt C)	40.4 %	87.5 %	mc
P0.5 cascaded-Γ 2-section	0.39 %	0.98 %	mc
P2.1 distributed-parameter (closed form)	$\mathbf{4.3\times10^{-6}}$ **%**	$\mathbf{2.7\times10^{-5}}$ **%**	mc

(`models/faultloc_distributed_param_model.py`); WP2.2 lands the closed-form analytic gradients (`inverse_estimation/faultloc_analytical_gradients.p` Appendix A §A.7).]

A. Section-model accuracy study (§ II.D)

The convergence study below has been updated under WP2.3 to include the closed-form distributed-parameter model from §IV. Three forward models are evaluated against a 50-section π reference (Saha-2010 half-π convention) on the $19\,\alpha \times 5 R_x$ operating grid at $f_0 = 50$ Hz; magnitude errors $\bar{\varepsilon}_H$ in percent.

Headline: the closed-form distributed-parameter model agrees with the 50-section π reference to better than $\mathbf{1}$ % across the operating grid (D-C acceptance). Empirically the agreement is at numerical precision, set by the 50-section reference's own sectioning floor (which we measured at $\sim\ 2 \times 10^{-5}$ % vs a 100-section reference — the residual is the WP1.3 lumped-line discretisation, not a real distributed-vs-π gap). The per-cell breakdown is in `outputs/phase2_reproduction.csv` (95 rows; `source_of_residual` column = `within_target` on every row).

Out of scope. Frequency-dependent line parameters ($R'(\omega)$ skin effect; earth-return $R'_g(\omega)$) are present in PSCAD's J. Marti FD line and EMTP-RV's FD model, but neither the P2.1 closed-form distributed model nor the WP1.3 50-section π reference includes them. The Phase-1 cross-platform validation (P1.4) showed that this lumped-vs-FD gap, at $f_0 = 50\,\text{Hz}$ with the short-feeder geometry studied here, is $< 1\,\%$ in $|H|$ across the grid (P1.3 finding); WP4.1 will quantify the impairment-class sensitivity. Frequency-dependent series impedance is therefore deferred to Phase 4 and treated as an *intentional* omission at the §IV / §V layer.

III. Transfer-Function Identification

[§ III analytical / DFT / cost - body migrates from v1. Among the impedance / spectral methods compared in § VII, [26] is the closest single-ended antecedent; the v1 manuscript's [2] and [10] both refer to this single entry (P0.3 de-duplication).]

IV. Two-Stage Optimiser

[§ IV body migrates from v1; WP0.4 sub-tasks 6/7/8 add global-optimum capture statistic, hyperparameter sensitivity sub-table, `tic/toc` CPU report.]

V. Simulation Setup and Noise Model

[§ V-A simulation programme, § V-B dual-channel noise model. All headline numerical values cited here MUST reuse the macros defined in the preamble - never hand-type $F_s = 10\,kHz$, $N_s = 200$, one cycle window or any of the headline percentages.]

VI. Results and Discussion

[§ VI body migrates from v1. Every numerical claim about location error, R_x error or the CRLB envelope reuses the macros defined in the preamble so values stay byte-identical with the abstract and the glossary in `docs/glossary.md`. Six panels are placed below; CRLB envelope overlays are added by WP1.6.]

Headline numerical findings (WP0.1 echo)

The two-stage estimator returns a mean fault-location error of $0.009\,\%$ noiseless and $1.18\,\%$ at $\text{SNR}_I = 20\,\text{dB}$ (see Glossary entry `LOC-ERR`). The mean arc-resistance error is $2.42\,\%$ noiseless and $\leq 8\,\%$ across $100 - 2000\,\Omega$ at SNR $\geq 30\,\text{dB}$ (Glossary `RX-ERR`). The empirical RMS estimator stays within $15\,\%$ of the Cramér–Rao Lower Bound at SNR $\geq 40\,\text{dB}$ (Glossary `CRLB-GAP`). All experiments use $F_s = 10\,\text{kHz}$, $N_s = 200$, one cycle window (Glossary `SAMP-CFG`).

VII. Comparison with Existing Methods

[§ VII Table 3-bis with four numerical competitors lands at WP4.5 (Phase 4): Paramo-2023 eigenvalue, Iurinic-2018 spectral, Cui-Weng-2020 micro-PMU, Zeng-2021 double-ended HIF.]

Figure placeholder – TF-magnitude modelling error of the N_s-section π-model versus a 50-section reference, swept over $N_s \in \{2, 5, 10, 20\}$ and $\alpha \in [0.05, 0.95]$ at $R_x = 1\,000\,\Omega$, $f_0 = 50\,\text{Hz}$.
(PDF: `outputs/fig_section_convergence.pdf`)

Fig. 1. Section-count convergence study. Horizontal axis: per-unit fault location α (dimensionless). Vertical axis: TF-magnitude modelling error vs the 50-section reference, in percent. $N_s = 2$ is the model used by the optimiser of § IV; deeper sectionalisation is treated as the reference. R_x is held at $1\,000\,\Omega$ and the source frequency at $f_0 = 50\,\text{Hz}$.

Figure placeholder – mean fault-location error vs current-channel SNR_I at fixed voltage-channel SNR_V, averaged across the 720-case grid.
(PDF: `outputs/fig_snr_sweep.pdf`)

Fig. 2. Mean fault-location error vs current-channel SNR_I (in dB; the rightmost cluster at 50 dB on the abscissa denotes the noiseless case). Vertical axis: mean per-cell relative location error in percent, on a logarithmic scale. Each curve corresponds to a fixed SNR_V level ($\{20, 30, 40, \text{noiseless}\}$ dB). The dominance of SNR_I over SNR_V at the same dB level is the empirical justification for the dual-channel noise model in § V.

Figure placeholder – mean fault-location error heatmap over the $\alpha \times R_x$ plane, averaged over the 4×4 noise grid per cell.
(PDF: `outputs/fig_alpha_rx_heatmap.pdf`)

Fig. 3. Mean fault-location error across the $9\,\alpha \times 5\,R_x$ grid, averaged over the $4\,\text{SNR}_V \times 4\,\text{SNR}_I$ noise classes. Horizontal axis: arc resistance R_x in ohms (logarithmic). Vertical axis: per-unit fault location α (dimensionless). Colour: $\log_{10}$ of the mean per-cell relative location error in percent; lighter shading indicates lower error. Near-source behaviour ($\alpha < 0.2$) is reported separately in § VIII as it is dominated by the CRLB floor rather than estimator behaviour.

VIII. Cramér–Rao Lower Bound Analysis

The bound used by the v1 manuscript treats H_{meas} as contaminated by circular complex Gaussian noise with constant variance. This linearisation is *not* self-consistent with the § V-B dual-channel AWGN noise model: $H_{\text{meas}} = I_{\text{bin}}/V_{\text{bin}}$ is the ratio of two independent complex Gaussians, whose density is the Marsaglia–Nadarajah–Pogány form [**?**], [27], [28], not Gaussian. The Gaussian-on-H linearisation is valid only in the asymptotic $|I| \gg \sigma_I$ regime — the opposite of the HIF regime, where $|H|^2\sigma_V^2$ becomes comparable to σ_I^2.

We therefore derive two corrected bounds (full derivation in Appendix B):

a) Proper-complex-Gaussian-ratio FIM.: At high SNR_V (Geary–Hinkley [27] $|V_{\text{phase}}|/\sigma_V > 4$, satisfied across the entire 720-case grid for time-domain $\mathrm{SNR}_V > -8\,\mathrm{dB}$) the ratio density admits a complex-Gaussian approximation with a *signal-dependent* per-component variance $\sigma_H^2 = (\sigma_I^2 + |H|^2\sigma_V^2)/|V_{\text{phase}}|^2$. The resulting FIM is $F_{kl}^{\text{proper}} = (1/\sigma_H^2)\,\mathrm{Re}[(\partial_{\theta_k} H)^*(\partial_{\theta_l} H)]$, implemented in `inverse_estimation/faultloc_crlb_proper.py`.

b) Joint dual-channel FIM (Nehorai–Hawkes, 2000) [29].: Working directly in $(V(t), I(t))$ time-domain space and projecting onto (α, R_x) via the chain rule with $\partial V/\partial\theta \equiv 0$ and $\partial I/\partial\theta = \mathrm{Re}[(\partial H/\partial\theta)V_{\text{phase}}e^{j\omega_0 t}]$ gives the ground-truth bound in waveform space, implemented in `inverse_estimation/faultloc_crlb_dualchannel.py`.

c) Cross-check.: The two FIMs share a common Fisher kernel and differ in their prefactor; their ratio is $F^{\text{proper}}/F^{\text{dual}} = \sigma_I^2/(\sigma_I^2 + |H|^2\sigma_V^2)$. At $\sigma_V \to 0$ (V perfectly known) the two are exactly equal; at finite σ_V the proper-ratio bound is looser by exactly the factor expected from the loss of information when only the H-ratio is observed. Both bounds are tested for consistency in `tests/test_crlb_consistency.py` (5 tests, all pass).

d) Four headline findings (WP1.6 corrected).:

1) The proper-ratio bound supersedes the v1 Gaussian-on-H linearisation; the latter under-estimates the true bound and is invalid in the HIF regime.
2) The dual-channel bound is tighter than the proper-ratio bound whenever $\sigma_V > 0$, by a factor that quantifies the information loss from working in H-ratio space rather than on the raw waveforms.
3) The empirical RMS estimator stays within 15 % of the Cramér–Rao Lower Bound at SNR $\geq$ 40 dB of the proper-ratio CRLB across the high-SNR cells of the 720-case grid (Glossary entry `CRLB-GAP`; overlay panel `outputs/phase1_crlb_overlay/`).
4) The Geary–Hinkley validity flag is satisfied across the full 720-case grid; the Gaussian-on-ratio approximation is therefore appropriate for the WP1.6 reporting envelope.

IX. Conclusion and Future Work

This paper presented a single-ended joint estimator of HIF location and arc resistance from a single power-frequency admittance bin under a dual-channel additive-white-Gaussian-noise model. Numerically, the estimator delivers a mean fault-location error of 0.009 % (noiseless) and 1.18 % at $\mathrm{SNR}_I = 20\,\mathrm{dB}$. The arc-resistance envelope is the principal scope statement of the estimator: $\leq 8\,\%$ across $100 - 2000\,\Omega$ at SNR $\geq 30\,\mathrm{dB}$. The empirical RMS error remains within 15 % of the Cramér–Rao Lower Bound at SNR $\geq 40\,\mathrm{dB}$.

A. Roadmap (12 / 24 / 36 months)

The future work below is anchored to Phases 1–5 of the SAMBPS DTaaS *HIF-TF Project Execution Plan v1.0* (`docs/FaultLocationIdentification_ExecutionPlan.pdf` § 2 and § 4).

12-month horizon (Phases 1–2, decision gates D1–D2).

1) Cross-platform validation on PSCAD, EMTP-RV and a 50-section pure-MATLAB reference, plus 100-trial Monte-Carlo per grid cell with 95th-percentile confidence intervals (Phase 1, WP1.1–WP1.5).
2) Re-derivation of the Fisher Information Matrix under the proper-complex-Gaussian-ratio framework (Kuruoğlu-2018) and the joint dual-channel FIM in (V, I) space (Nehorai-Hawkes-2000), with the Geary-Hinkley validity test reported per cell (Phase 1, WP1.6).
3) Continuously parametrised distributed-parameter $H(j\omega_0; \alpha, R_x)$ via cascaded ABCD / Bergeron blocks, retiring the 39.44 % two-section modelling-error ceiling, with analytical $\partial H/\partial\alpha, \partial H/\partial R_x$ replacing the central-difference gradient (Phase 2, WP2.1–WP2.5).

24-month horizon (Phases 3–4, decision gates D3–D4).

4) Generalisation to three-phase $Y_{abc}(j\omega_0)$ on the IEEE 13-, 34- and 123-node test feeders with laterals, tap loads and at least one distributed generator (Phase 3, WP3.1–WP3.4).
5) First-order Taylor–Fourier phasor estimator (Platas-Garza-2010) replacing the single-bin DFT, plus Hermann–Krener / STRIKE-GOLDD identifiability mapping of $J(\alpha, R_x)$, and a multi-port FIM extension (Phase 3, WP3.5–WP3.6).
6) Field-grade impairment classes (impulsive, harmonic, CT/PT saturation per IEEE C37.110, $\pm 0.5\,\mathrm{Hz}$ off-nominal per IEEE C37.118.1 P-class, ADC quantisation) and arc-class diversification (Cassie–Mayr–Kizilcay, Wang-2020 distortion-controllable, Torres-2022 stochastic), plus a numerical Table 3-bis benchmarking four competitor methods on identical 720-case waveforms (Phase 4, WP4.1–WP4.5).

36-month horizon (Phase 5, decision gate D5).

7) RTDS / Typhoon HIL pilot at IIT-Madras with real relay-class IED integration via IEC 61850-9-2 SV; live HIF arc scenarios targeting end-to-end estimate latency below five power-frequency cycles (Phase 5, WP5.1–WP5.3).
8) Productisation as the SAMBPS DTaaS Protection-Validation module v1.0 with REST + Python client API and identifiability + CRLB overlays surfaced to the operator (Phase 5, WP5.4).
9) Second journal paper (IEEE TPWRD or TSG) bringing in HIL evidence, the four-competitor numerical benchmark and the multi-port CRLB (Phase 5, WP5.5).

B. Integration into the SAMBPS DTaaS Protection-Validation module

The estimator's single-frequency, training-free, single-ended operation maps cleanly onto the SAMBPS

DTaaS scenario-engine: the digital twin streams synthetic dual-channel V/I waveforms through the locator, surfaces the identifiability flag from `adaptation/faultloc_identifiability_check.py`, and overlays the corrected CRLB envelope on the operator's location estimate. This binds the estimator to a maintainable validation harness rather than a one-shot study, and forms the deployment vehicle for the v1.0 release at decision gate D5 (see Execution Plan § 4.6 WP5.4 and § 11 KPI K16).

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